

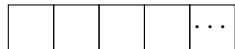
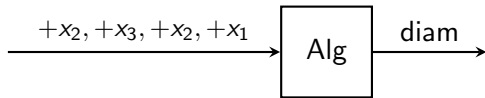
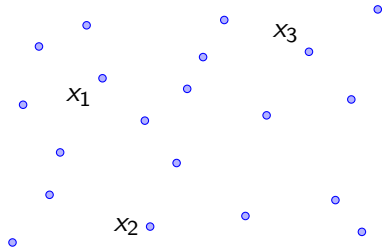
A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams

Sanjeev Khanna, Ashwin Padaki, Krish Singal, and Erik Waingarten

October 6, 2026

Streaming Diameter

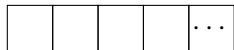
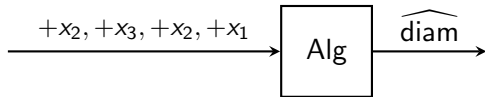
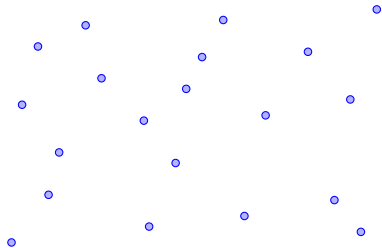
Metric Space (M, d)



Frequency Vector x

Streaming Diameter

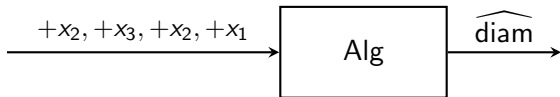
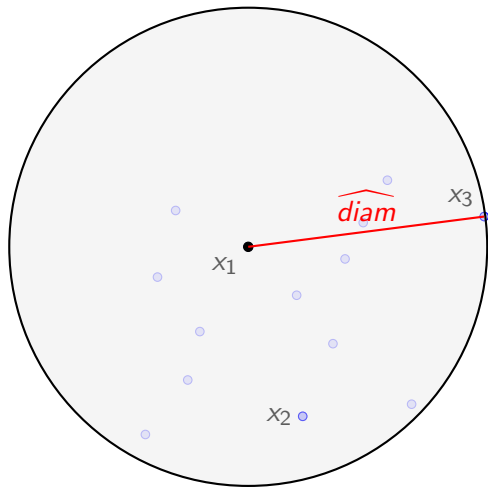
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Warm-Up: Insertion-Only Streams

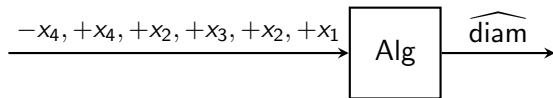
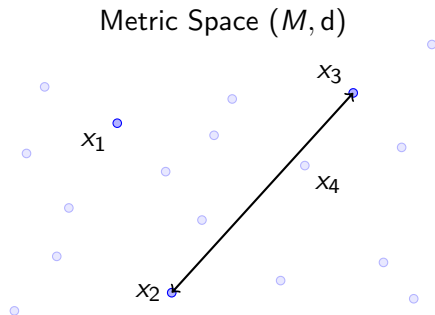
Metric Space (M, d)



1	2	1	0	...
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Frequency Vector x

Dynamic Streams



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Frequency Vector x

Dynamic Streams

But what about dynamic streams?

But what about dynamic streams?

- Insertion-only algorithm finds a surviving point and computes the radius around it
- But it's unclear how to identify such a surviving point a priori

Intuition: Streaming Diameter in ℓ_∞

When can we say something non-trivial?

Intuition: Streaming Diameter in ℓ_∞

When can we say something non-trivial?

There exists a deterministic and exact dynamic streaming algorithm for $([\Delta]^d, \ell_\infty)$ using space $O(d \cdot \Delta)$

1. For each coordinate, maintain a histogram for each value
2. Diameter is the maximum difference along a coordinate

Streaming Diameter in General Metrics

Theorem (Matousek)

Any metric space (M, d) can be embedded into ℓ_∞^k with distortion $2c - 1$ and $k = \tilde{O}(n^{1/c})$.

This immediately implies a dynamic streaming algorithm for c -approximate diameter in general metric spaces using $n^{O(1/c)} \cdot \Delta$ space.

Theorem

There exists a dynamic streaming algorithm for c -approximate diameter in general metric spaces using $n^{O(1/c)} \cdot \log \Delta$ space

Streaming Diameter in General Metrics

What about lower bounds?

Streaming Diameter in General Metrics

What about lower bounds?

- Euclidean MST [CJLW21, CCAJ⁺23]: $n^{O(1/\sqrt{c})}$ space
- Euclidean Facility Location [CJK⁺22]: $n^{O(1/c)}$ space
- EMD over $[\Delta]^2$: $\Delta^{O(1/c)}$
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Motivating Question: Can we show a polynomial space lower bound for Euclidean diameter?

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Motivating Question: Can we show a polynomial space lower bound for Euclidean diameter?

Answer: Not yet :(But we can show the existence of a metric on which diameter is hard

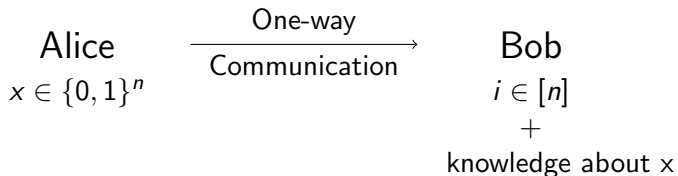
Streaming Diameter in General Metrics

Theorem (Informal Main Result)

There exists a metric space (M, d) where any dynamic streaming algorithm c -approximating diameter requires $n^{\Omega(1/c)}$ space.

Towards a Lower Bound via Communication Complexity

Indexing with a Twist



Bob must output x_i

Towards a Lower Bound via Communication Complexity

Alice

Bob

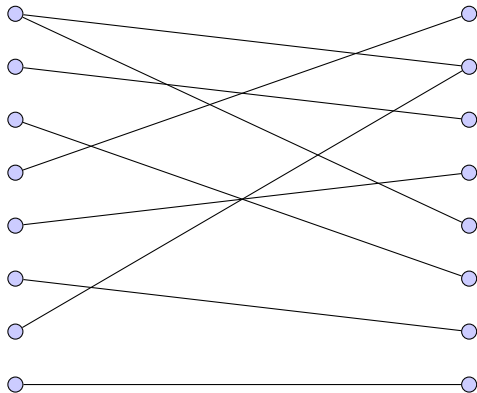
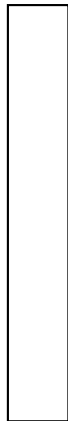
$$x \in \{0, 1\}^n$$

U

$$G : d_G(u_i, v_i) \geq 2c$$

V

$$y \in \{*, 0, 1\}^n$$



$$i \in [n]$$



i

$N(v_i)$

Towards a Lower Bound via Communication Complexity

- This communication game looks a lot like Indexing but with a *twist*
- Studied as *Index Coding with Side Information*
- Its communication complexity remains open, but progress has been made when restricting to linear protocols over finite fields

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Results generalize when domain is $\mathbb{F}$ and linear protocols are over $\mathbb{F}$, but will need a different approach to show *general* streaming lower bounds

Min Rank

Consider the following incomplete matrix induced by a graph $G = (U, V, E)$

- 1's on the diagonal
- $M_{ij} = 0$ if $(u_i, v_j) \in E$

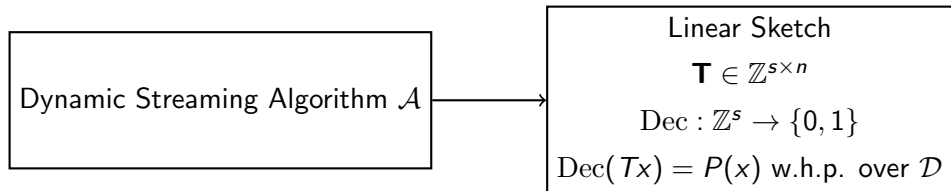
$$M = \begin{bmatrix} 1 & & & & & & & & \\ & 1 & & & & & & & \\ & & 0 & & & & & & \\ & & & 1 & & & & & \\ & & & & 0 & & & & \\ & & & & & 1 & & & \\ & & & & & & 0 & & \\ & & & & & & & 1 & \\ & & & & & & & & 0 & \\ & & & & & & & & & 1 \end{bmatrix}$$

$$\text{Minrank}(G) := \min_{\text{Real completions } M' \text{ of } M} \text{rank}(M')$$

Streaming Algorithms are Equivalent to Linear Sketches

[Gan08, LNW14]

For a decision problem P and a distribution $\mathcal{D}$ over frequency vectors $x \in \{0, \dots, m\}^n$.



$$s \lesssim \text{Space}(\mathcal{A}) \cdot \log m$$

A Naive Plan

Fix a *hard* distribution $\mathcal{D}$ over frequency vectors $x \in \{0, \dots, m\}^n$.

Dynamic Streaming Algorithm $\mathcal{A}$



Linear Sketch

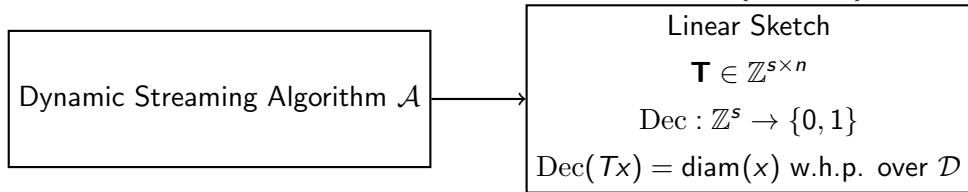
$$\mathbf{T} \in \mathbb{Z}^{s \times n}$$

$$\text{Dec} : \mathbb{Z}^s \rightarrow \{0, 1\}$$

$$\text{Dec}(\mathbf{T}x) = \text{diam}(x) \text{ w.h.p. over } \mathcal{D}$$

A Naive Plan

Fix a *hard* distribution $\mathcal{D}$ over frequency vectors $x \in \{0, \dots, m\}^n$.



Find a matrix $H \in \mathbb{R}^{n \times s}$ such that HT is a minrank completion of some $\mathbf{G} \sim \mathcal{G}(n, n, p)$.

$$\text{Space}(\mathcal{A}) \cdot \log m \gtrsim s = \text{rank}(T) \geq \text{rank}(HT) \geq \tilde{\Omega}(np)$$

The Problem

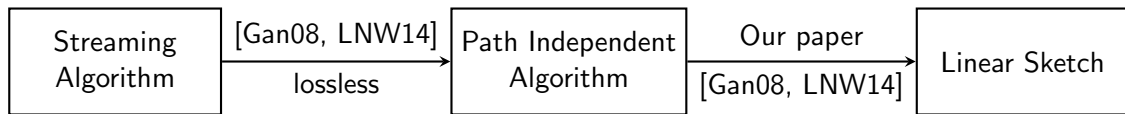
But there's a problem!

- When choosing our *hard* distribution and constructing our matrix H , our argument requires $m \gg \exp(s)$
- This will give a vacuous lower bound on $\text{Space}(\mathcal{A})$!

$$\text{Space}(\mathcal{A}) \gtrsim \frac{s}{\log m}$$

Two Reductions

(scale-invariant functions)
 $f(ax) = f(x)$ for $a \in \mathbb{Z}^+$



(general functions,
 $\log m$)

Open Questions

1. Can we get a similar result for Euclidean space?
2. Where else can we apply the *loss/less* reduction from linear sketches to streaming algorithms for scale-invariant functions?

Thank you :)

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